

Integrated Computation Offloading, UAV Trajectory Control, and Resource Allocation Against Jamming in SAGIN

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Abstract—In this paper, we study the computation offloading problem against an active attacker in space-air-ground integrated networks (SAGIN), where joint optimization of partial computation offloading, unmanned aerial vehicle (UAV) trajectory control, computation and resource allocation is performed. Our design aims to minimize the maximum computation time of individual tasks among ground users while satisfying energy consumption constraints. To tackle the underlying non-convex optimization problem, we use the alternating optimization approach to iteratively solve three sub-problems, namely, partial offloading control and bit allocation over time slots, computation resource and bandwidth allocation, and UAV trajectory control, until convergence. Furthermore, the successive convex approximation method is employed to solve the non-convex sub-problems and improve the resilience of the SAGIN against active attacks. Via extensive numerical studies, we illustrate the effectiveness of our proposed design compared to baselines under the effect of an active attacker.

Index Terms—SAGIN, UAV trajectory, mobile edge computing, jamming, non-convex optimization.

I. INTRODUCTION

For future wireless communications networks, a number of promising technologies are being considered, including satellite communications, unmanned aerial vehicle (UAV) communications, and mobile edge computing (MEC) [1]. Satellite communications, especially those utilizing low Earth orbit (LEO) satellites, offer notable advantages due to their low propagation delay, high communication rate, and their capacity to provide uninterrupted communication services across vast geographical regions [2]. This becomes particularly valuable in scenarios where terrestrial base stations are either absent or incapacitated due to natural disasters. To this end, by leveraging the complementary strengths of space, air, and ground networks, space-air-ground integrated networks (SAGIN) have emerged as an effective means of providing high-quality and ubiquitous communications [3].

Computation offloading and resource allocation in the SAGIN have attracted great attention [4]–[6] where computation tasks can be offloaded from ground users (GUs) to UAV-mounted edge servers and satellites to save GUs' energy and/or improve computation latency. Specifically, minimization of the maximum delay experienced by different GUs by jointly optimizing UAV-device association, task assignment, power control, bandwidth allocation, computation resource, and UAV placement was studied in [4]. The authors in [5] considered a multi-user MEC system and optimized user association and task partitioning to achieve minimum average latency for all

GUs where independent and dependent sub-tasks are explored. In addition, a SAGIN setting with a single satellite, a single UAV, and multiple small cells was studied in [6] whose design aims to maximize the sum rate of the small cells by jointly optimizing the user association, sub-channel and power allocation considering the maximum delay constraint.

Moreover, due to the broadcast nature and openness of SAGIN, several researchers have investigated the malicious jamming attacks [7]–[9]. In particular, the joint transmit power, UAV trajectory, and cooperative jamming design to achieve a positive secrecy rate of the system was considered in [7], meanwhile the anti-jamming satellite-to-UAV communication problem with the presence of the aerial jammer and the cross-tier interference was investigated in [8]. Furthermore, the authors in [9] considered a coordinated satellite-UAV communication system, where the UAV performs the environmental reconnaissance task with the assistance of satellites in a hostile jamming environment. However, the computation offloading under the jamming attacks still has not been considered in these studies.

To the best of our knowledge, none of the existing work has studied the UAV trajectory control with computation offloading and resource allocation against an active attack targeting the availability of the system, such as jamming, in SAGIN. We study the computation offloading problem against jamming in SAGIN through joint optimization of partial computation offloading, UAV trajectory control, and computation and resource allocation. This design aims to minimize the maximum computation time of individual tasks among GUs while satisfying energy consumption constraints. To tackle the underlying non-convex optimization problem, we use the alternating optimization approach. Moreover, the successive convex approximation (SCA) method is employed to solve the computation resource and bandwidth allocation and UAV trajectory control sub-problems. Numerical results are presented to show the impacts of different parameters and the importance of trajectory optimization on the achievable min-max processing time of individual tasks among GUs, showing the resilience of the proposed approach in the presence of active attackers.

II. SYSTEM MODEL

We study the computation offloading design in the SAGIN illustrated in Fig. 1 where the terrestrial network layer provides services for K GUs, the aerial network layer employs a UAV, and the space network layer relies on a LEO satellite for connections to a distant cloud server. In addition, a malicious attacker (such as a jammer) attempts to interrupt legitimate

This work was supported in part by the Fonds de Recherche du Québec–Nature et Technologies (FRQNT) and in part by the Regroupement Stratégique en Microsystemes du Québec (ReSMiQ).

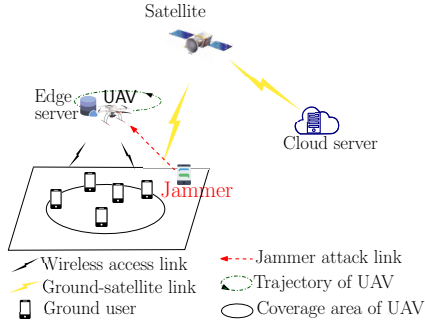


Fig. 1: Proposed system model with anti-jamming in SAGIN.

GU-UAV communications and it can jam the full bandwidth of the network.

Partial computation offloading is considered where each GU partitions its computation task into three sub-tasks to be processed locally, offloaded and processed at the UAV-mounted edge server and at the cloud server, respectively [4], [5]. This network model reflects practical scenarios such as natural disasters or temporary events where the base station is damaged or far away from the considered network area.

We denote the set of GUs as $\mathcal{K} = \{1, \dots, K\}$. All GUs are assumed to have fixed horizontal coordinates of $\mathbf{g}_k^0 = (x_k^0, y_k^0)$, $\forall k \in \mathcal{K}$. The attacker (jammer) is also located on the ground at $\mathbf{g}^j = (x^j, y^j)$. Besides, we assume that the UAV flies at a fixed altitude H with a flight period of $T > 0$, divided into N time slots, denoted as set $\mathcal{N} = \{1, \dots, N\}$. In addition, uplink transmissions from GUs to UAV are assumed using the frequency division multiple access (FDMA). The uplink bandwidth, B , is partitioned into orthogonal sub-bands. We assume that the GU-UAV and GU-satellite associations are fixed during the computation process. We also neglect the download time of the computation results.

A. Computation Task Model

We consider that each GU k has a latency-critical task denoted by $U_k = (f_k, s_k, c_k, T_k^{\max})$, where f_k denotes the computation demand (CPU cycles/sec), s_k is the size of input raw data, c_k (CPU cycles/bit) is the computation resource required for 1-bit input data, and $T_k^{\max}$ is the maximum permissible latency of the task U_k . Hence, the task processing time is:

$$T_k = \max \left\{ T_k^{\text{lo}}, T_k^{\text{ed}}, T_k^{\text{cl}} \right\}, \quad (1)$$

where T_k^{lo} , T_k^{ed} , and T_k^{cl} represent the task execution (and possibly offloading) time at the GU, UAV-mounted edge server, and cloud server, respectively. Hence, the processing time for GU k must satisfy the delay constraint $T_k \leq T_k^{\max}$. Moreover, we introduce variables α_k^{lo} and α_k^{ed} , ($0 \leq \alpha_k^{\text{lo}}, \alpha_k^{\text{ed}} \leq 1$) that represent the fractions of input data to be processed locally and at the UAV, respectively. Hence, $(1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{ed}})$ represents the fraction of input data from GU k to be processed at the distant cloud server.

B. UAV Trajectory Control

We assume the UAV's energy is sufficiently large to cover its flight operation and wireless communications over the

flight period T . The horizontal coordinates of UAV in time slot n is denoted as $\mathbf{u}[n] = (x^u[n], y^u[n])$. We assume that each UAV must come back to its initial position at the end of the flight period, i.e., $\mathbf{u}[1] = \mathbf{u}[N]$. The trajectory of the UAV must meet the following constraints: $\|\mathbf{u}[n+1] - \mathbf{u}[n]\|^2 \leq D_{\max}^2$, $n = 1, \dots, N-1$, where $\|\cdot\|$ denotes the Euclidean norm, $D_{\max} \triangleq V_{\max} \delta$ is the maximum horizontal distance that the UAV can travel in one time slot with the length of $\delta = \frac{T}{N}$.

C. Computing Models

The task execution time and energy consumption are discussed in the following.

1) *Local Computing Model*: The task execution time can be expressed as

$$T_k^{\text{lo}} = \frac{\alpha_k^{\text{lo}} s_k c_k}{f_k}. \quad (2)$$

It also must satisfy the delay requirement $T_k^{\text{lo}} \leq T_k^{\max}$. The energy consumption due to local task execution will be

$$E_k^{\text{lo}} = \kappa \alpha_k^{\text{lo}} s_k c_k (f_k)^2, \quad (3)$$

where κ is the effective switched capacitance depending on the chip architecture [10].

2) *UAV-Mounted Edge Server Model*: Let $\tau_k^u[n]$ denote the number of offloading bits from the GU k to the associated UAV over the time slot n . Besides, let us denote the computing capacity of the UAV to handle the sub-task offloaded from the GU k at time slot n by $f_k^u[n]$ (CPU cycles/second). Then, the task execution time at the UAV in time slot n can be computed as $T_k^{\text{edc}}[n] = \frac{\tau_k^u[n] c_k}{f_k^u[n]}$.

In addition, we assume that the communication links from the ground to the UAV are dominated by line-of-sight (LoS) propagation, where the channel quality is mostly dependent on the distances. Moreover, the channel power gain from GU k to UAV in time slot n is assumed to follow the free-space path loss model, and it can be expressed as $g_k[n] = \frac{\rho_0}{H^2 + \|\mathbf{u}[n] - \mathbf{g}_k^0\|^2}$, where ρ_0 presents the channel power gain at the reference distance of 1 m. Hence, the achievable rate of the uplink transmission from GU k to UAV in time slot n denoted by $R_k^u[n]$ can be expressed as

$$R_k^u[n] = \beta_k^u[n] \log_2 \left(1 + \frac{P_k^u g_k[n]}{P^j g^j[n] + \beta_k^u[n] \sigma^2} \right), \quad (4)$$

where $\beta_k^u[n]$ and P_k^u represent the bandwidth allocated to GU k in time slot n and the transmit power of GU k for its uplink transmission, respectively. In addition, P^j and $g^j[n]$ represent the transmit power of the jammer and channel power gain from the jammer to UAV, which can be calculated as $g^j[n] = \frac{\rho_0}{H^2 + \|\mathbf{u}[n] - \mathbf{g}^j\|^2}$. Moreover, σ^2 denotes the power density of the additive white Gaussian noise (AWGN) at the receiver. Then, the transmission time from GU k to UAV in time slot n for the data related to the offloaded sub-task can be expressed as $T_k^{\text{edt}}[n] = \frac{\tau_k^u[n]}{R_k^u[n]}$. The corresponding energy consumption can be calculated as

$$E_k^{\text{edt}}[n] = \frac{\tau_k^u[n] P_k^u}{R_k^u[n]}. \quad (5)$$

Moreover, we assume that the partial task from GU k is offloaded and processed completely at the UAV in each time slot. Then we have the following constraints

$$T_k^{\text{ed}}[n] = \left(\frac{\tau_k^u[n]c_k}{f_k^u[n]} + \frac{\tau_k^u[n]}{R_k^u[n]} \right) \leq \delta, \forall k, n. \quad (6)$$

3) *Satellite Cloud Computing Model*: We ignore the processing time at the cloud server and the cloud energy consumption involved. Hence, the total processing and propagation time from GU k to the cloud server can be calculated as:

$$T_k^{\text{cl}} = (1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{ed}})s_k \left(\frac{1}{R_k^s} + \frac{1}{R^{\text{cl}}} \right) + T_k^{\text{prop}}, \quad (7)$$

where R_k^s, R^{cl} stand for the transmission rates between the GU k and the satellite and between the satellite and the cloud server, respectively, T_k^{prop} represents the total propagation delay. Moreover, the energy consumption of GU k for transmitting the data to the satellite can be calculated as

$$E_k^s = \frac{(1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{ed}})s_k P_k^s}{R_k^s}, \quad (8)$$

where P_k^s denotes the transmit power of GU k to the satellite.

D. Problem Formulation

The total energy consumption of GU k can be expressed as

$$E_k = E_k^{\text{lo}} + \sum_{n \in \mathcal{N}} E_k^{\text{ed}}[n] + E_k^s, \forall k. \quad (9)$$

For convenience, we gather different decision variables such as partial offloading control $\mathbf{A} = \{\alpha_k^{\text{lo}}, \alpha_k^{\text{ed}}, \forall k\}$, bit allocation $\mathbf{\Upsilon} = \{\tau_k^u[n], \forall k, n\}$, bandwidth allocation $\beta = \{\beta_k^u[n], \forall k, n\}$, computation resource allocation $\mathbf{F} = \{f_k^u[n], \forall k, n\}$, and UAV trajectory control $\mathbf{U} = \{\mathbf{u}[n], \forall n\}$.

The task processing time T_k in (1) is a non-linear function with respect to $\mathbf{A}, \mathbf{\Upsilon}, \beta, \mathbf{F}$, and $\mathbf{U}$. We introduce the function $\eta(\mathbf{A}, \mathbf{\Upsilon}, \beta, \mathbf{F}, \mathbf{U}) = \max_{k \in \mathcal{K}} T_k$. Then, our problem becomes equivalent to minimizing $\eta(\mathbf{A}, \mathbf{\Upsilon}, \beta, \mathbf{F}, \mathbf{U})$, which is more tractable. It can be formulated as:

$$(\mathbf{P}): \min_{\mathbf{A}, \mathbf{\Upsilon}, \beta, \mathbf{F}, \mathbf{U}} \eta \quad (10)$$

$$\text{s.t. } T_k^{\text{lo}} \leq \eta, T_k^{\text{cl}} \leq \eta, \sum_{n \in \mathcal{N}} T_k^{\text{ed}}[n] \leq \eta, \forall k, \quad (10a)$$

$$T_k^{\text{lo}} \leq T_k^{\text{max}}, T_k^{\text{cl}} \leq T_k^{\text{max}}, \sum_{n \in \mathcal{N}} T_k^{\text{ed}}[n] \leq T_k^{\text{max}}, \forall k, \quad (10b)$$

$$T_k^{\text{ed}}[n] \leq \delta, \forall k, n, \quad (10c)$$

$$E_k \leq E_k^{\text{max}}, \forall k, \quad (10d)$$

$$\sum_{n \in \mathcal{N}} \tau_k^u[n] = \alpha_k^{\text{ed}} s_k, \forall k, \quad (10e)$$

$$\sum_{k \in \mathcal{K}} \beta_k^u[n] \leq B, \forall n, \quad (10f)$$

$$\sum_{k \in \mathcal{K}} f_k^u[n] \leq F^{\text{max}}, \forall n, \quad (10g)$$

$$\mathbf{u}[1] = \mathbf{u}[N], \|\mathbf{u}[n+1] - \mathbf{u}[n]\|^2 \leq D_{\text{max}}^2, n=1, \dots, N-1, \quad (10h)$$

$$0 \leq \alpha_k^{\text{lo}}, \alpha_k^{\text{ed}}, 1 - \alpha_k^{\text{lo}} - \alpha_k^{\text{ed}} \leq 1, \forall k, \quad (10i)$$

$$\beta_k^u[n], f_k^u[n], \tau_k^u[n] \geq 0, \forall k, n, \quad (10j)$$

where constraints (10a)-(10c) capture the delay requirements for the GUs. Constraints (10f) capture the bandwidth allocation while (10g) present the UAV's computation constraints where F^{max} denotes the maximum computation resource of UAV. It can be seen that the objective and constraint functions (10a)-(10c) are non-linear. Hence, problem (10) is non-convex, which is difficult to solve optimally. In the following section, we propose to compute good feasible solutions by dividing the problem into sub-problems and using the alternating optimization method.

III. PROPOSED OPTIMIZATION APPROACH

A. Optimization of Partial Offloading Control and Bit Allocation Over Time Slots Sub-Problem

Given $\{\mathbf{F}, \beta, \mathbf{U}\}$, the sub-problem optimizing the partial offloading control and bit allocation can be formulated as

$$(\mathbf{P1}): \min_{\mathbf{A}, \mathbf{\Upsilon}} \eta \quad (11)$$

s.t. (10a) – (10e), (10i), (10j).

It can be verified that the objective function and all constraints are linear with respect to $\mathbf{A}$ and $\mathbf{\Upsilon}$. Therefore, problem (11) is a linear problem (LP) and it can be solved by using the CVX-Gurobi solver.

B. Optimization of Computation Resource and Bandwidth Allocation Sub-Problem

Given $\{\mathbf{A}, \mathbf{\Upsilon}, \mathbf{U}\}$, the sub-problem optimizing the computation resource and bandwidth allocation is still non-convex. We first introduce auxiliary variables

$$\xi_k[n] = R_k^u[n] = \beta_k^u[n] \log_2 \left(1 + \frac{C_k[n]}{D_k[n] + \beta_k^u[n]} \right), \quad (12)$$

where $C_k[n] = \frac{P_k^u g_k[n]}{\sigma^2}$ and $D_k[n] = \frac{P^i g^j[n]}{\sigma^2}$.

It can be verified that $\beta_k^u[n] \log_2 \left(1 + \frac{C_k[n]}{D_k[n] + \beta_k^u[n]} \right)$ is a concave function with respect to $\beta_k^u[n]$ ¹. Using the successive convex approximation (SCA) method, the upper-bound for this concave function by using the first-order Taylor expansion at the given point $\beta_k^{u,r}[n]$ in the r -th iteration of the approximation process can be derived as

$$\begin{aligned} \xi_k[n] &\leq \beta_k^{u,r}[n] \log_2 \left(1 + \frac{C_k[n]}{D_k[n] + \beta_k^{u,r}[n]} \right) + \\ &\quad \left(\log_2 \left(1 + \frac{C_k[n]}{D_k[n] + \beta_k^{u,r}[n]} \right) - \right. \\ &\quad \left. \frac{\log_2(e) C_k[n] \beta_k^{u,r}[n]}{(D_k[n] + \beta_k^{u,r}[n])(C_k[n] + D_k[n] + \beta_k^{u,r}[n])} \right) (\beta_k^u[n] - \beta_k^{u,r}[n]) \\ &\triangleq R_k^{\text{ub}}[n]. \end{aligned} \quad (13)$$

Then, this sub-problem of optimizing the computation resource and bandwidth allocation can be approximated by the following problem:

¹The Hessian of function $f(x) = x \log_2(1 + a/(b+x))$ can be derived as $\nabla^2 f(x) = -\frac{\log_2(e) a(a(2b+x) + 2b(b+x))}{(b+x)^2(a+b+x)^2} < 0, \forall x, a, b > 0$. Therefore, $f(x)$ is a concave function with respect to x .

$$(\mathbf{P2}): \min_{\mathbf{F}, \beta, \Xi} \eta \quad (14)$$

$$\text{s.t.} \sum_n \left(\frac{\tau_k^u[n]c_k}{f_k^u[n]} + \frac{\tau_k^u[n]}{\xi_k[n]} \right) \leq \eta, \forall k, \quad (14a)$$

$$\sum_n \left(\frac{\tau_k^u[n]c_k}{f_k^u[n]} + \frac{\tau_k^u[n]}{\xi_k[n]} \right) \leq T_k^{\max}, \forall k, \quad (14b)$$

$$\left(\frac{\tau_k^u[n]c_k}{f_k^u[n]} + \frac{\tau_k^u[n]}{\xi_k[n]} \right) \leq \delta, \forall k, n, \quad (14c)$$

$$E_k^{\text{lo}} + E_k^{\text{s}} + \sum_n \frac{\tau_k^u[n]P_k^u}{\xi_k[n]} \leq E_k^{\max}, \forall k, \quad (14d)$$

constraints (10f), (10g), (10j), (13),

where $\Xi = \{\xi_k[n], \forall k, n\}$.

Since $\frac{1}{f_k^u[n]}$ and $\frac{1}{\xi_k[n]}$ are convex functions with respect to $f_k^u[n]$ and $\xi_k[n]$, respectively, problem (14) is convex, which can be solved effectively by using the CVX-Gurobi solver.

C. Optimization of UAV Trajectory Sub-Problem

Given $\{\mathbf{A}, \mathbf{\Upsilon}, \mathbf{F}, \beta\}$, the sub-problem optimizing $\mathbf{U}$ is still non-convex. We solve it by using the SCA method. To approximate this problem, we introduce auxiliary variables $\gamma_k[n] = R_k^u[n]$, $S_k^{\text{all}}[n] \geq S_k^0[n] + S^j[n]$, $\Pi_k^0[n] \leq H^2 + \|\mathbf{u}[n] - \mathbf{g}_k^0\|^2$, and $\Pi^j[n] \leq H^2 + \|\mathbf{u}[n] - \mathbf{g}^j\|^2$. The approximation processes are similar to that in [11]. Therefore, this problem can be approximated by:

$$(\mathbf{P3}): \min_{\mathbf{U}, \Gamma, \mathbf{S}, \Pi} \eta \quad (15)$$

$$\text{s.t.} \sum_n \left(\frac{\tau_k^u[n]c_k}{f_k^u[n]} + \frac{\tau_k^u[n]}{\gamma_k[n]} \right) \leq \eta, \forall k, \quad (15a)$$

$$\sum_n \left(\frac{\tau_k^u[n]c_k}{f_k^u[n]} + \frac{\tau_k^u[n]}{\gamma_k[n]} \right) \leq T_k^{\max}, \forall k, \quad (15b)$$

$$\left(\frac{\tau_k^u[n]c_k}{f_k^u[n]} + \frac{\tau_k^u[n]}{\gamma_k[n]} \right) \leq \delta, \forall k, n, \quad (15c)$$

$$E_k^{\text{lo}} + E_k^{\text{s}} + \sum_n \frac{\tau_k^u[n]P_k^u}{\gamma_k[n]} \leq E_k^{\max}, \forall k, \quad (15d)$$

$$\gamma_k[n] \leq R_k^{\text{lb}}[n], \forall k, n, \quad (15e)$$

$$\Omega_k^{0,r}[n] \leq P_k^u \rho_0, \forall k, \quad (15f)$$

$$\Omega_k^{j,r}[n] \leq P^j \rho_0, \quad (15g)$$

$$S_k^0[n] + S^j[n] \leq S_k^{\text{all}}[n], \forall k, n, \quad (15h)$$

$$\begin{aligned} \Pi^j[n] &\leq \|\mathbf{u}^r[n] - \mathbf{g}^j\|^2 + \\ &2(\mathbf{u}^r[n] - \mathbf{g}^j)^T(\mathbf{u}[n] - \mathbf{u}^r[n]) + H^2, \forall n, \end{aligned} \quad (15i)$$

$$\begin{aligned} \Pi_k^0[n] &\leq \|\mathbf{u}^r[n] - \mathbf{g}_k^0\|^2 + \\ &2(\mathbf{u}^r[n] - \mathbf{g}_k^0)^T(\mathbf{u}[n] - \mathbf{u}^r[n]) + H^2, \forall k, n, \end{aligned} \quad (15j)$$

constraints (10h),

where $\Gamma = \{\gamma_k[n], \forall k, n\}$, $\mathbf{S} = \{S_k^{\text{all}}[n], S^j[n], S_k^0[n], \forall k, n\}$, $\Pi = \{\Pi^j[n], \Pi_k^0[n], \forall k, n\}$, $\Omega_k^{0,r}[n] = \frac{1}{4}[(S_k^0[n] + \Pi_k^0[n])^2 + (S_k^{0,r}[n] - \Pi_k^{0,r}[n])^2 - 2(S_k^{0,r}[n] - \Pi_k^{0,r}[n])(S_k^0[n] - \Pi_k^0[n])]$, $\Omega_k^{j,r}[n] = \frac{1}{4}[(S^j[n] + \Pi^j[n])^2 + (S^{j,r}[n] - \Pi^{j,r}[n])^2 - 2(S^{j,r}[n] - \Pi^{j,r}[n])(S^j[n] - \Pi^j[n])]$, $R_k^{\text{lb}}[n] =$

Algorithm 1 Integrated Partial Offloading, Computation, Bandwidth Allocation, and UAV Trajectory Control Algorithm

Require: $\mathcal{K}, B, T$, locations of GUs, satellites and cloud server;

1: **Initialization:** $\Upsilon^0, \mathbf{A}^0, \mathbf{F}^0, \beta^0, \mathbf{U}^0$;

Ensure: Min-max processing time (η); Let $r = 1$;

2: **repeat**

3: Solve sub-problem (11) to obtain Υ^r and $\mathbf{A}^r$;

4: Solve sub-problem (14) to obtain β^r and $\mathbf{F}^r$;

5: Solve sub-problem (15) to obtain $\mathbf{U}^r$;

6: Update $r = r + 1$;

7: **until** Convergence

8: Return $\eta^*, \Upsilon^*, \mathbf{A}^*, \mathbf{F}^*, \beta^*, \mathbf{U}^*$.

TABLE I: Simulation Parameters

Parameter	Description	Value
K	Number of GUs	[10 – 20]
T	Flight period	10 s
δ	Length of each time slot	0.5 s
H	Altitude of UAV	100 m
P_k^u	Transmit power of GU with UAV	0.3 W
P_s^s	Transmit power of GU with satellite	0.5 W
P^j	Transmit power of jammer	[0 – 1] W
c_k	Computation resource required for 1-bit input data (CPU cycles/bit)	300
$E_k^{\max}$	Maximum available energy of GU k	1 J
f_k	Computation resource of GU (CPU cycles/s)	2×10^8
$F^{\max}$	Maximum computation resource of UAV (CPU cycles/s)	3×10^9
σ^2	Noise power	-174 dBm/Hz
κ	Effective switched capacitance	10^{-28}
R_k^s	Transmission rate from GU to satellite	10 Mbps
R^{cl}	Transmission rate from satellite to cloud server	10 Mbps
$T_k^{\max}$	Maximum delay tolerance of GU k	4 s
T_k^{prop}	Total propagation delay from GU k to cloud server	20 ms
$V_{\max}$	Maximum speed of UAV	50 m/s
B	Total available bandwidth	[50 – 100] MHz

$$\beta_k^u[n](\log_2(S_k^{\text{all}}[n] + \beta_k^u[n]\sigma^2) - \log_2(S^{j,r}[n] + \beta_k^u[n]\sigma^2) - \frac{\log_2(e)}{S^{j,r}[n] + \beta_k^u[n]\sigma^2}(S^j[n] - S^{j,r}[n])).$$

Since $\frac{1}{\gamma_k[n]}$ is a convex function with respect to $\gamma_k[n]$, problem (15) is convex, which can be solved effectively by using the CVX-Gurobi solver.

D. Integrated Algorithm

Using the results above, problem (P) can be solved by using an integrated algorithm based on the alternating optimization method as described in Algorithm 1.

IV. NUMERICAL RESULTS

We consider different scenarios with a far cloud server from a network area with natural disasters or temporary events. The parameter setting for our simulations is done similarly to that in [4], [10], [11] and the values of key parameters are summarized in Table I. Specifically, we consider a network area of 1000m $\times$ 1000m, and the jammer is located at the center with the horizontal coordinate of (500, 500). Moreover, the task size values s_k are set randomly in the range of [1, 10] Mbits. Furthermore, the initial values of partial offloading control variables are randomly generated in $\alpha_k^{\text{lo}}, \alpha_k^{\text{ed}} \in [0, 0.5]$, and a uniform allocation of bit, computation resource, and bandwidth is applied, i.e., $\tau_k^u[n] = \alpha_k^{\text{ed}} s_k / N$, $f_k^u[n] = F^{\max} / K$, and $\beta_k^u[n] = B / K$, respectively.

To investigate the effectiveness of the proposed algorithms, we consider the following baselines. In the “baseline algorithm”, we initially set circular UAV’s trajectory to serve the

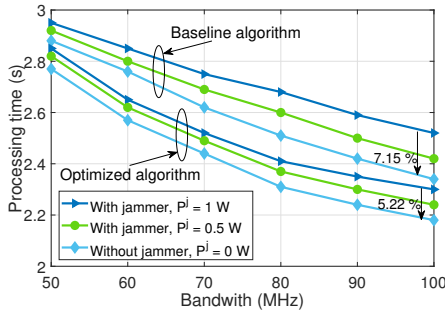


Fig. 2: Processing time for different bandwidth values.

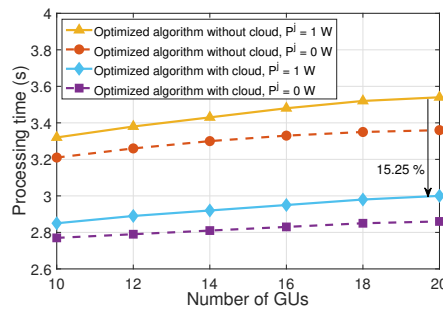


Fig. 3: Processing time for different number of GUs.

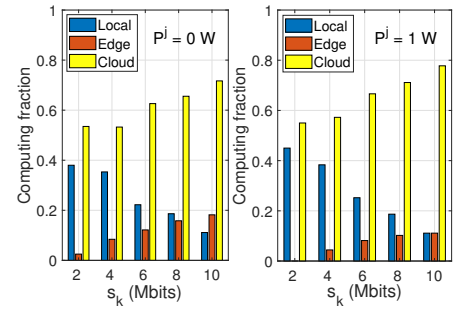


Fig. 4: Computation distribution with different task size values.

corresponding groups of GUs. The values of partial offloading control variables are randomly set and a uniform allocation of bit, computation resource, and bandwidth is applied. For comparison, the “optimized algorithm” represents our proposed design where all variables are optimized.

We first compare the effectiveness of our proposed algorithm in terms of processing time with the baselines in Fig. 2. This figure considers 10 GUs and $P^j = [0, 0.5, 1]W$. It can be seen that the processing time for individual tasks decreases as the total bandwidth becomes larger. This is reasonable because larger bandwidth allows larger transmission rates and hence larger fractions of data to be offloaded and processed at the more resourceful edge servers, achieving processing time savings for both the edge servers and GUs. Moreover, it can also be seen that our proposed design can save processing time compared to the baseline algorithm. Specifically, the gap between the scenarios with and without jammer attacks in our optimized algorithm (5.22%) is smaller than those with the baseline algorithm (7.15%). These results indicate the effectiveness of our proposed algorithm in solving the problem of computation offloading, UAV trajectory control, and resource allocation against jamming in SAGIN.

Fig. 3 shows the processing time with different numbers of GUs in the scenario of $B = 50\text{MHz}$ and $P^j = [0, 1]W$. As expected, the processing time for individual tasks increases as the number of GUs increases. This is because, given the bandwidth and computation resources, the transmission rates of GUs decrease with a larger number of served GUs. Hence, it takes more time to process completely the sub-tasks at the local, edge, or cloud server. Also, we see that the design with the jammer results in a higher processing time, and the design with a cloud server reduces the processing time. For instance, it could reduce 15.25% in the case of 20 GUs. Moreover, the gap between the scenarios with and without jammer attacks in our proposed design is smaller than those in the scenarios without a cloud server.

Furthermore, Fig. 4 shows the computation load distribution over network layers in the scenarios of 10 GUs and $B = 50\text{MHz}$ with and without jammer attacks, i.e., $P^j = 1W$ and $P^j = 0W$, respectively. This figure shows that larger task size values lead to less computation load distributed at the GUs, while the scenarios without jammer attacks result in a higher computation load to be processed at the edge servers.

V. CONCLUSION

In this paper, we have studied the joint optimization of partial offloading control and bit allocation, computation resource and bandwidth allocation, and UAV trajectory control against jamming for the SAGIN. We have employed the alternating optimization approach to solve the underlying problem and leveraged the SCA method to tackle the non-convex computation resource and bandwidth allocation and UAV trajectory control sub-problems. Numerical results have demonstrated the effectiveness of the proposed algorithm compared to the baselines and the strong effects of various system parameters such as the numbers of GUs and bandwidth as well as the data size values to computation load distribution and processing time for individual GUs’ tasks.

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